DIMITRIOS GOUSOPOULOS



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Home address: 14 Dervenakion Street, Agia Paraskevi

Date of birth: 5 September 1987

Military service: completed

Profile

Researcher and university lecturer focusing on digital technologies, Artificial Intelligence, data analysis and digital application development. My research and teaching connect LLM-based systems, media data, digital culture and digital transformation with computer science, statistics and data analysis.

Education and Postdoctoral Research

Postdoctoral Research in Artificial Intelligence and Quantitative Methods, 2025 - present, **Panteion University of Social and Political Sciences**

> **Research topic:** "Development and application of statistical and econometric models for training executives of organizations and businesses in Artificial Intelligence technologies."

PhD in Educational Technology, 2012-2016, National and Kapodistrian University of Athens. **Grade: 10/10**

> **Doctoral thesis:** "Educational Approaches to the Special Theory of Relativity Using Digital Technologies."

The doctoral thesis investigates how interactive digital media can support the understanding of complex scientific concepts. It includes the design and development of interactive simulations using HTML5, CSS3, JavaScript and jQuery, with an emphasis on information visualization, interface design and active manipulation of parameters by users. The research combines qualitative investigation of conceptual difficulties with quantitative evaluation of an educational intervention incorporating the simulations, linking educational technology with interactive application design and the study of learning through digital media.

> **SYLFF Fellow** (Sasakawa Young Leaders Fellowship Fund), **Nippon and Tokyo Foundations**
>
> Doctoral thesis in the National Archive of PhD Theses (EKT)

MSc in Science Education, 2010-2012, National and Kapodistrian University of Athens. **Grade: 9.79/10**

BSc in Physics, 2005-2010, National and Kapodistrian University of Athens. **Grade: 7.45/10**

Teaching Experience in Higher Education

Independent Course Teaching

Adjunct Lecturer in Computer Science, Department of International and European Economic Studies, Athens University of Economics and Business, spring semester 2025

Independent teaching: *Computer Applications II (undergraduate elective course)*

Adjunct Lecturer in Statistics and Computer Science, Department of Public Administration, Panteion University of Social and Political Sciences, spring semester 2024

Independent teaching of the following courses:
Statistics I (compulsory undergraduate course)
Computer Applications - Data Analysis (undergraduate elective course)

Postgraduate Teaching and Co-teaching

Co-instructor in Quantitative Methods, MSc in Economics, Department of Public Administration, Panteion University of Social and Political Sciences, academic year 2026-2027.

Adjunct Lecturer in Computer Science, Department of Communication, Media and Culture / Department of International, European and Area Studies, Panteion University of Social and Political Sciences, 25 December 2024 - present

> *Co-teaching of* ***Economy 4.0****, including the unit* ***Artificial Intelligence: the driving force behind Economy 4.0***
> *and teaching of* ***Bootcamp: Intro to the Digital Era & Research Methods***
>
> *within the interdepartmental MSc programme* ***Digital Transformation: e-Diplomacy, e-Campaigning and Digital Law (Digi-DCL)***

Laboratory Instructor for the postgraduate course "Computer Applications in Education", Department of Education, National and Kapodistrian University of Athens. **Project contracts:** 7 January-4 February 2013, 19 April-31 July 2013 and 17 March-30 June 2014.

Designed and delivered postgraduate training in digital technologies and their integration into teaching practice to enhance learning outcomes.

Research Projects and Funded Research in Higher Education

Researcher, Host Institution: Panteion University, 1 January-30 June 2026

Project: "Methodology for Monitoring and Quality Assurance of Radio Audience Measurement in Attica"
Funding body: Civil Company for Radio Audience Measurement (AEMAR). **Special Account for Research Funds project code: 1512**
Role: Researcher - development of radio audience measurement methodology, statistical and analytical models, and data quality assurance
Responsibilities:

- Development and updating of a standard methodology for radio audience measurement
- Use of AI technologies, statistics and analytical models
- Development of telephone sampling frames and data pipelines
- Data quality and reliability assessment

University Lifelong Learning Programmes

Instructor, "Artificial Intelligence in Education", Centre for Training and Lifelong Learning (KEDIVIM), Ionian University. Project contract: 1 July-26 August 2025

Taught units on the use of Artificial Intelligence tools in educational practice, computational thinking, and the social and ethical dimensions of AI in education.

Scientific and Professional Experience in Policy and Research Bodies

Paid scientific and professional work in structured projects and programmes following the award of the doctoral degree

Assessment Item Writer and Reviewer, Graded Difficulty Item Bank, Institute of Educational Policy (IEP), 2021-2023
Project: "Graded Difficulty Item Bank for Secondary Education (General and Vocational Upper Secondary Schools)", **MIS code 5070818**
Role: Member of the Special Scientific Team of Assessment Item Writers and Reviewers (subject: Physics)
Responsibilities

- Writing, scientific checking, revision and evaluation of assessment items of graded difficulty before their inclusion in the Item Bank.
- Checking scientific validity, clarity and suitability of content against the prescribed specifications and assessment criteria.
- Participation in a structured process of scientific evaluation, revision and quality assurance of educational content.
- Contribution to the development and quality enhancement of a national digital repository of educational content through the IEP information system for managing and submitting assessment items.

Content Developer and Evaluator of Open Educational Resources (OERs), Learning Scenarios and Digital Courses, Institute of Educational Policy (IEP), August 2024 - June 2026
Project: "Transformation of conventional curricula and educational content into open-source, interactive digital environments accessible to all, with integrated Artificial Intelligence", **MIS code 5174140**
Role: Expert - Developer and Evaluator of digital educational content (subject: Physics in Vocational Upper Secondary Education)
Responsibilities

- Preparation of specifications for OERs, learning scenarios and digital courses as a basis for developing interactive digital content
- Scientific evaluation of detailed specifications, OERs and completed digital courses
- Assessment of the educational validity, scientific accuracy and quality of digital deliverables
- Quality control of digital deliverables and content/data quality assurance
- Preparation of evaluation reports and participation in feedback and revision cycles through an information system
- Participation in a project for the digital transformation of educational content with integrated Artificial Intelligence

Research and Advisory Collaborations with Institutes

Researcher, Research Institute for Tourism (RIT / ITEP). Project contract: 1 October 2026 - 31 August 2027

- *Collection, processing and statistical analysis of research data, preparation of scientific reports and presentation of research findings.*
- *Integration of Artificial Intelligence technologies into the Institute's scientific and research processes.*
- *Design, organization and delivery of educational activities.*

Training Consultant on the Use of Artificial Intelligence Applications, Research Institute for Tourism (RIT / ITEP). Project contract: 1 May-30 June 2025

- *Training the Institute's research team in the use of AI tools and their application to its research, report preparation, and presentation development and formatting. Contribution to the design of an AI platform covering the full information lifecycle of the Institute's research.*

Executive Training in Artificial Intelligence

Artificial Intelligence Instructor, TheSigmaEducation / Sigma Business Network, July 2026

Delivered ISO/IEC 42001 Foundation seminars on Artificial Intelligence management systems as part of executive training for public- and private-sector organizations. Participants included officials from the Special Secretariat for Artificial Intelligence and Data Governance of the Ministry of Digital Governance.

Teaching Experience in Secondary Education

Physics Teacher, Athens A Directorate of Secondary Education, 2025 - present

Head of Physics, Lower and Upper Secondary Education, Erasmios Greek-German School, 2024-2025

Physics Teacher, Lower and Upper Secondary Education, Erasmios Greek-German School, 2018-2025

Scientific Activities in Physics Competitions and Olympiads

Member of the Scientific Committee for the Panhellenic Physics Competitions, Association of Greek Physicists and University of Athens, 2013-2016

Writing, revising and evaluating questions for national competitions organized by the Association of Greek Physicists and the University of Athens.

Member of the Training Team for Greek Delegations to the International Physics Olympiads, Association of Greek Physicists and University of Athens, 2011-2016

Training students in advanced theoretical and experimental physics; designing teaching activities and laboratory exercises for the preparation of national teams.

Books

Physics for the First Year of Lower Secondary School, Student Textbook, 2013 [in Greek]
Diophantus Publishing - Ministry of Education, ISBN 978-960-06-4416-6

Physics for the First Year of Lower Secondary School, Teacher's Book, 2014 [in Greek]
Diophantus Publishing - Ministry of Education, ISBN 978-960-06-4568-2

Physics for the Final Year of Upper Secondary School, 2015 [in Greek]
Othisi Publications, ISBN 978-960-415-356-5

Peer-reviewed Publications in International Journals

1. Gousopoulos, D., Petrakos, G. (2026). **Exploring the Use of LLMs as Educational Tools in Cultural Heritage Education.** *Communications in Computer and Information Science*, vol 2838, pp. 461–471. Springer, Cham. https://doi.org/10.1007/978-3-032-15069-1_33
2. Gousopoulos, D. (2026). **AI-Generated Simulations in STEM Education: A Scoping Review.** *Journal of Digital Educational Technology*. Accepted for publication. Manuscript ID: JDET-46792-2026.
3. Gousopoulos, D. (2026). **Evaluating a GPT-Powered Tutor in Secondary Science Education Using a Fuzzy Comprehensive Approach.** *Discover Education*. Springer Nature. https://doi.org/10.1007/s44217-026-01913-z
4. Gousopoulos, D. (2026). **Teacher Acceptance of AI-Generated Educational Simulations: A Fuzzy DEMATEL Causal Analysis.** *EIKI Journal of Effective Teaching Methods*, vol 4(2). https://doi.org/10.59652/43y3xd28
5. Gousopoulos, D. (2026). **Can AI-Assisted Inquiry Enhance Students' Decision-Making Skills in Socio-Scientific Issues? A Three-Group Experimental Study on Climate Change.** International Journal of Advanced Multidisciplinary Research and Studies vol 6(3). https://doi.org/10.62225/2583049X.2026.6.3.6516
6. Gousopoulos, D. (2025). **Learning-by-coding as an educational method of approaching projectile motion**. *The Physics Teacher*, *63(6)*, *488–491.*
7. Gousopoulos, D. (2024). **Educational approach of Special Relativity using a custom GPT as teaching assistant.** *EIKI Journal of Effective Teaching Methods*, *2(4)*.

Peer-reviewed Publications in Greek Journals

1. Athanasakopoulou, S., Gousopoulos, D., & Petrakos, G. (2026). **Artificial Intelligence in Higher Education: Perceptions, Concerns and Use of AI Tools among Public Administration Students.** *Dialogues in Public Administration* [in Greek; forthcoming].
2. Gousopoulos, D. (2024). **The Special Theory of Relativity in Education.** *Neos Paidagogos*, March, 107-114 [in Greek].
3. Gousopoulos, D. (2024). **Fast and Slow Thinking and Science Education.** *Neos Paidagogos*, January, 28-36 [in Greek].

Peer-reviewed International Conference Publications

1. Gousopoulos, D., Petrakos, G. (2026). **Smart Cultural Heritage: Design and Development of an AI-Powered Mobile Application for Personalized Museum Exploration.** 5th International Conference, TMM_CH 2026, Athens (forthcoming).
2. Gousopoulos, D. (2026). **Factors Influencing Teachers' Acceptance of AI-Generated Simulations: A Fuzzy DEMATEL Analysis.** 11th International Conference on Education and Culture in the 21st Century, Athens [in Greek].
3. Gousopoulos, D., Kapotis, E., & Kalkanis, G. (2015). **Students' difficulties in understanding the basic principles of Relativity after standard instruction**. *ESERA Conference*, Helsinki, Finland.
4. Gousopoulos, D., Oikonomidis, S., & Kalkanis, G. (2010). **A hands-on experimentation and educational study for a 2000 years-old puzzle, the Mpemba Effect**. *HSci 2010 – 7th International Conference on Hands-on Science*, University of Crete, Rethymno.

Peer-reviewed Greek Conference Publications

1. Gousopoulos, D. (2025). **Prompt Engineering in Science Teaching.** *14th Panhellenic Conference on Science Education*, Thessaloniki [in Greek]. https://doi.org/10.12681/codiste.9891
2. Kapotis, E., Tsakonas, P., & Gousopoulos, D. (2017). **Learning Objects for the Simulation of Physical Phenomena.** *10th Panhellenic Conference on Science Education*, Crete [in Greek].
3. Gousopoulos, D., & Kalkanis, G. (2015). **Difficulties in Understanding the Special Theory of Relativity.** *9th Panhellenic Conference on Science Education*, Thessaloniki [in Greek].

4. Gousopoulos, D., & Kapotis, E. (2015). **Relativistic and Classical Limits Using Interactive Software.** *9th Panhellenic Conference on Science Education*, Thessaloniki [in Greek].
5. Kalkanis, G., Gkikopoulou, O., Kapotis, E., & Gousopoulos, D. (2015). **Using "Interstellar" in Science Teaching.** *9th Panhellenic Conference on Science Education*, Thessaloniki [in Greek].
6. Kalkanis, G., Gkikopoulou, O., Kapotis, E., Gousopoulos, D., et al. (2015). **Experiments and Hands-on Constructions in Physics.** *9th Panhellenic Conference on Science Education*, Thessaloniki [in Greek].
7. Gkikopoulou, O., Kapotis, E., Gousopoulos, D., & Kalkanis, G. (2013). **Building a Motor and a Generator for Educational Experimentation.** *8th Panhellenic Conference on Science Education*, Volos [in Greek].
8. Kalkanis, G., Gkikopoulou, O., Imvrioti, D., Kapotis, E., & Gousopoulos, D. (2013). **Investigating and Discovering in Primary Education.** *8th Panhellenic Conference on Science Education*, Volos [in Greek].
9. Gousopoulos, D., & Katsiampoura, G. (2012). **Byzantine Natural Philosophy: Michael Psellos.** *7th Panhellenic Conference on the History and Philosophy of Science* [in Greek].

Manuscripts under Review

1. Gousopoulos, D. (2026). **Fuzzy Comprehensive Evaluation of ChatGPT Acceptance among Digital Transformation Postgraduate Students**. *Journal of Applied Research in Higher Education* (Emerald), manuscript JARHE-01-2026-0019.R2. Revised manuscript resubmitted 26 September 2026; under review.

Preprints

1. Gousopoulos, D. (2026). **Code to Learn with Generative AI: A Theoretically Grounded Framework for Artifact Construction in Upper-Secondary Education**. *Preprints.org*. https://doi.org/10.20944/preprints202606.1139.v1
2. Gousopoulos, D. (2025). **Exploring Teachers' Perceptions of ChatGPT Through Prompt Engineering**. *arXiv*:2510.08634.
3. Gousopoulos, D. (2024). **Developing a Custom GPT Based on Inquiry-Based Learning for Physics Teachers**. *arXiv*:2412.18617.
4. Gousopoulos, D. (2024). **Enhancing Physics Teachers' Self-Concept with Generative AI-Driven Educational Scenarios**. *TechRxiv* preprint.
5. Gousopoulos, D. (2024). **Thinking Fast and Slow and Science Education**. *arXiv*:2402.05497.
6. Gousopoulos, D. (2023). **Investigating Students' Scientific Reasoning Through Heuristic and Analytical Thought Processes**. *arXiv*:2311.07158.

Research Metrics and Online Research Profiles

Google Scholar: https://scholar.google.com/citations?user=qLZJdUYAAAAJ&hl=el
GitHub: https://github.com/dimgouso

Digital Applications, Artificial Intelligence and Computational Systems

- **AI-powered applications:** Development of applications based on LLMs and AI analytics, including **Diplomat Coach**, an LLM-based negotiation training system, and **Hospitality Intelligence Tool**, an AI-driven analytics and decision-support tool for hotels.
- **Interactive web applications and simulations:** Development of **7 interactive applications using HTML, CSS and JavaScript** and **2 simulations using Python**, with empirical evaluation of their use in learning environments.
- **Research computing projects:** Development of Python tools for multicriteria decision-making, fuzzy methods, decision support and simulations, including **Agent Decision Intelligence (ADI)** and **QA-FTOPSIS**.

Technical and Methodological Skills

AI & Analytics
LLMs, AI tutoring systems, predictive modeling, Fuzzy Methods, PLS-SEM (SmartPLS), learning analytics

Computational Tools
Python (Pandas, NumPy, Scikit-Learn), R, SPSS, SQL, Javascript (simulations), GitHub, Colab

Languages

English: Proficient, CEFR C2 **French:** Upper-intermediate, CEFR B2